

HAOS-IIP Magnon Address Stability: Testable Signatures

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Scope

This note creates the parallel spin-sector prediction layer. It is not a new spin-wave theory, not an external magnon proof of HAOS-IIP, and not a claim that ferrons and magnons share a literal substrate.

Calibration and Boundary

The committed calibration source is Materials Bridge Line A: ferron persistence proxy `0.957192`, mean spectral recoverability `0.928455`, 15/15 target-window peaks near `3.13 THz`, and raw STFT-grid status `NO_STFT_DATA_FOUND`.

A user-provided alpha-Fe₂O₃ magnon audit anchor reports persistence proxy `0.902` and raw public STFT-grid boundary `NO_PUBLIC_STFT_GRID_FOUND`. The corresponding audit artifact was not found in this repo at note creation time, so `0.902` is recorded as a user-provided anchor until the audit files are committed.

Candidate External Anchors

Candidate audit targets include Li et al., “Anisotropic Coherent Propagation of Sub-Terahertz Magnons in Altermagnetic alpha-Fe₂O₃,” *Advanced Optical Materials* 2026; Yang et al., “Spin-orbit torque manipulation of sub-terahertz magnons in antiferromagnetic alpha-Fe₂O₃,” *Nature Communications* 2024; Sun et al., “Observation of Chiral Magnon Band Splitting in Altermagnetic Hematite,” *Physical Review Letters* 2025; Choe et al., “Long-lived zone-boundary magnons in an antiferromagnet,” *Nature Communications* 2025; and Yuan et al., “Inelastic Neutron Scattering Study of Field Dependence of Magnetic Excitations in CoTiO₃,” *Physical Review B* 2024. These references motivate target datasets; they do not validate HAOS metrics by themselves.

Field-Facing Predictions

Prediction	Testable readout	Falsifier
Crystal-orientation-dependent narrowband magnon address	Sub-THz exchange magnons should show highest coherence and narrowest linewidth only for specific wavevector / crystal-axis alignments.	If orientation scans show isotropic broadband response, or aligned records do not outperform misaligned controls, the prediction fails.
Thickness / boundary-dependent group velocity	Magnon packet velocity should scale with thickness, confinement, or boundary condition while target-window recoverability remains above 0.70 before visible propagation failure.	If controlled thickness or boundary series show no systematic velocity scaling and no distance-ordered delay beyond uncertainty, the prediction fails.
Target-address recovery after perturbation	After a secondary pulse, field quench, strain perturbation, or spin-orbit torque below damage threshold, the readout should return to the original target frequency window with recoverability at least 0.85.	If perturbation repeatedly causes complete thermalization or a different dominant mode with no return to the original address, the prediction fails.
Hybrid magnon-phonon / cross-sector locking	In vdW magnetic hybrids or coupled heterostructures, compatible magnon and phonon / polar modes should produce enhanced lifetime, cleaner linewidth, or more directional propagation than uncoupled controls.	If hybridization gives no lifetime, linewidth, directionality, or recoverability benefit beyond uncoupled controls, the prediction fails.

Boundary

These are field-facing diagnostic candidates. They do not claim new physics, a universal harmonic grid, or a replacement for spin-wave theory. The next required step is to commit the $\alpha\text{-Fe}_2\text{O}_3$ audit artifact or build a new magnon data bridge that loads public spectra / traces and runs the same bounded recoverability audit used for ferrons.